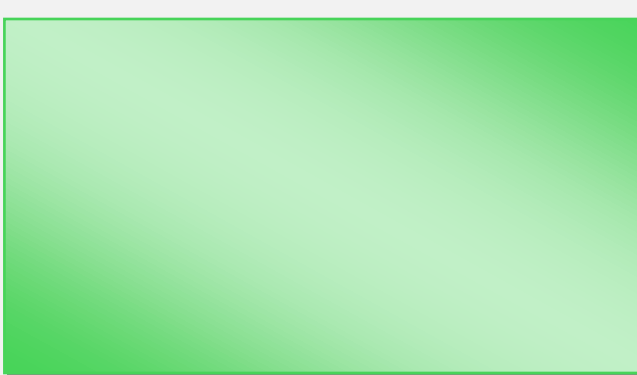




CARE4FOOD: CHOICES AND PERCEPTIONS
OF FOOD SECURITY AND SUSTAINABILITY IN
A CLIMATE CHANGE CONTEXT IN
PORTUGAL.

Dashboard descriptor



Care4Food Dashboard: Data, Methods, and Policy-Relevant Outputs

1. Overview and purpose

The Care4Food dashboard is a prototype collaborative, spatially explicit monitoring tool designed to support the analysis of food consumption, food environments, and sustainable diet behaviours in Portugal. It integrates citizen-generated behavioural data with spatial indicators to provide policy-relevant insights at municipal and NUTS III levels.

The dashboard was developed within the research project *Food choices and perceptions of food security and sustainability in the context of climate change in Portugal*, funded by the Social Observatory of the “la Caixa” Foundation through the Flash Call *Social Impacts of Climate Change* (FP23-2B073). The project responds to growing evidence that food-system transitions require territorially differentiated policies grounded in behavioural, perceptual, and spatial data.

Rather than producing static indicators, the Care4Food system enables continuous updating of sustainability signals at the local level, supporting benchmarking, cross-territorial learning, and exploratory policy analysis. The dashboard also represents a first step toward the creation of a Food Sustainability Observatory for Portugal.

2. Conceptual framework

The Care4Food dashboard is grounded in a territorial approach to food-system sustainability, recognising that food choices and perceptions are shaped by the interaction between individual behaviours, food environments, and broader socio-spatial contexts. It explicitly addresses the gap between perceptions of dietary sustainability and observed practices, which has been identified as a key barrier to effective food-policy interventions.

By aggregating individual data at municipal and regional scales, the dashboard supports the analysis of spatial heterogeneity in sustainable food practices, revealing differences between coastal and inland areas, urban and rural municipalities, and territories with distinct socio-economic profiles. These spatial patterns are exploratory and participation-dependent, but they illustrate the added value of citizen-generated data for territorial food-policy design.

3. Data sources

The Care4Food dashboard integrates multiple data sources:

3.1 Citizen-generated survey data

The core dataset is a national survey on food habits conducted in Portugal in 2024. The survey collects information on: - Consumption of animal-based and plant-based proteins - Consumption of ultra-processed foods - Food purchasing profiles - Perceptions of food sustainability and food security

Survey responses are anonymised and aggregated for spatial analysis. Individual-level data are not publicly disclosed.

3.2 Contextual and spatial data

To contextualise behavioural data, the dashboard integrates external datasets, including: - OpenStreetMap (OSM): locations of food retail outlets (supermarkets, grocery stores, markets, fast-food and convenience stores) - INE (Statistics Portugal): administrative boundaries and socio-demographic context - DGT (Directorate-General for Territory): land-use data for excluding non-residential areas in spatial analyses.

4. Indicator construction and methods

4.1 Sustainable food score

A sustainable food score is calculated at the individual level based on: - Balance between animal and plant-based protein consumption - Intake of ultra-processed foods - Food purchasing behaviours.

Scores are aggregated by municipality and NUTS III region. Aggregated scores are dynamic: when a new participant completes the survey, municipal and regional values are automatically updated.

4.2 Food environment indicators

Food environment indicators include: - Fresh food store density - Fast-food and convenience store density

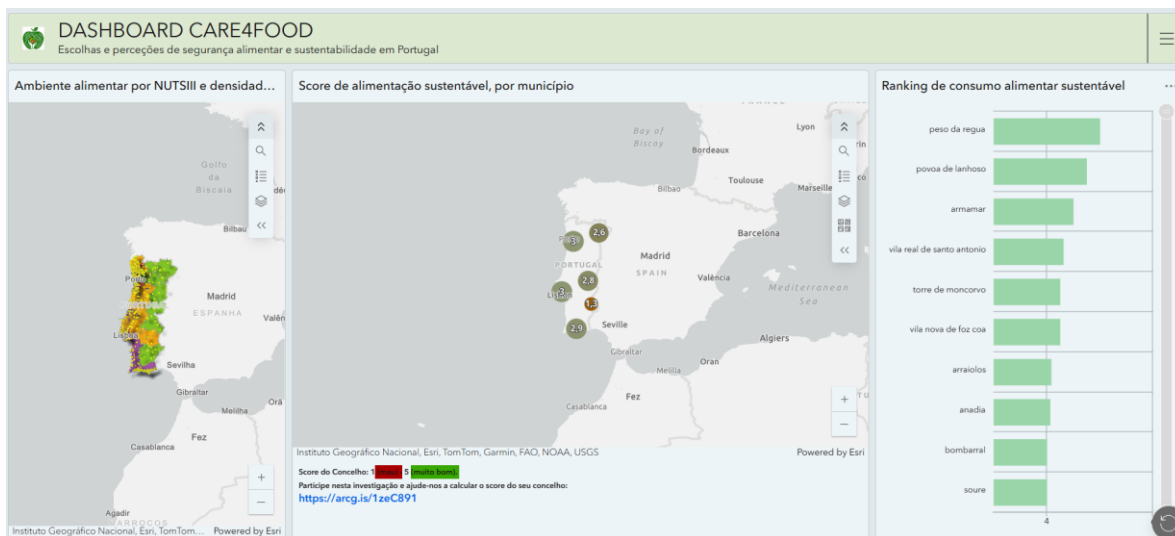
These indicators are built using kernel density estimation with a 500-metre radius, representing an approximate five-minute walking distance (extendable to fifteen minutes). Non-residential land-use areas are excluded to better approximate everyday food-access conditions.

More information is found at : ABRANTES, P., Marques da Costa, E., Cláudia M. Viana, & Eduardo Jonas Gomes. (2025). Care4Food: Choices and perceptions of food security and sustainability in a climate change context in Portugal (v0.2). Zenodo. <https://doi.org/10.5281/zenodo.17580594>

5. Spatial aggregation and visualisation

The dashboard presents data at two main territorial levels: - Municipal level - NUTS III regional level

Interactive maps and charts allow users to explore spatial disparities, compare territories, and identify patterns related to urbanisation, deprivation, and food access. While current participation levels limit inferential spatial analysis, the system is explicitly designed to scale as participation increases.



6. Policy relevance and applications

The Care4Food dashboard supports spatially explicit policy analysis in several ways:

1. **Information for decision-making:** By identifying geographic variation in perceptions and practices, the dashboard highlights where misperceptions may undermine policy effectiveness.
2. **Benchmarking and learning:** Municipal aggregation enables cross-territorial comparison, supporting the identification of priority domains and good practices.
3. **Cross-sectoral integration:** By combining behavioural, environmental, and spatial data, the system supports policy integration across health, environment, education, and spatial planning.

Potential applications include school food programmes, food-literacy initiatives, local food systems development, sustainable mobility to food outlets, and the articulation of food and health policies.

7. Toward a Food Sustainability Observatory

While the Care4Food dashboard does not yet constitute a formal observatory, it provides a clear pathway toward one. Its robust database, real-time dashboards, citizen-science features, and capacity for periodic data collection offer a scalable foundation for a Food Sustainability Observatory operating at municipal and regional levels.

Such an observatory would support evidence-based policymaking, foster civic participation, and enhance transparency and accountability in food-system transformation.

8. Limitations and future development

Current results are exploratory and dependent on participation levels. As the number and diversity of respondents increase, the system will enable more robust spatial comparisons and hypothesis testing, including the relationship between food environments, policy interventions, and perception–practice gaps.

Future developments include: - Expansion of participation - Integration of additional contextual indicators - Enhanced feedback mechanisms for participants - Strengthened interoperability with national and EU food-policy frameworks

9. Access to the live dashboard

The live version of the Care4Food dashboard is available at:
<https://www.arcgis.com/apps/dashboards/9088fa801b454f32afa39d5c84e10c7b>

Data availability

Individual-level survey data are not publicly available due to privacy and ethical considerations. Aggregated indicators are provided for research and policy-analysis purposes on-demand: patricia.abrantes@edu.ulisboa.pt.

Methodological notes

- Sustainable food scores are based on dietary composition and purchasing behaviour
- Food environment indicators use kernel density estimation (500 m radius)
- Non-residential land-use areas are excluded from accessibility calculations

Funding

Fundação “la Caixa” – Social Observatory Flash Call *Social Impacts of Climate Change* (FP23-2B073)

Suggested citation

Abrantes, P., et al. (2025). *Care4Food dashboard: Food consumption and sustainable diet scores in Portugal* (Version 1.0) [Data set]. Zenodo.
<https://doi.org/10.xxxx/zenodo.xxxx>

Contact

For further information, please contact the Care4Food research team.
patricia.abrantes@edu.ulisboa.pt.